

FMC

A program to manage, classify, cluster and plot
focal mechanism data

VERSION 1.10, SEPTEMBER 2026

José A. Álvarez-Gómez
Faculty of Geology. Universidad Complutense de Madrid



This text is distributed under the Creative Commons License:
Attribution-NonCommercial-ShareAlike 4.0 International ([CC BY-NC-SA 4.0](https://creativecommons.org/licenses/by-nc-sa/4.0/))

Contents

I The manual	5
1 Introduction	5
2 Versions	5
3 Installation	8
4 Usage	9
4.1 Input	10
4.2 Output	11
4.3 Plot	16
4.4 Clustering	22
4.5 Working with GMT	28
4.6 Command-line reference	29
4.7 Troubleshooting and known issues	30
5 License	30
II The background	31
6 Introduction	31
7 The focal mechanism	32
8 Parameters computed by FMC	34
9 Diagram	37
10 Focal Mechanism Classification	39
10.1 Comparison between rake-based and SMT axes-based clas- sifications	40
11 The source-type diagram	43
References	45

Acknowledgments

I have been using different versions of this program during the last decade. Initially I reworked some of the [Gasparini and Vannucci \(2003\)](#) FORTRAN subroutines on Matlab in order to obtain all the focal mechanism parameters from the Harvard CMT `psmeca` formatted catalog. During my research on seismotectonics I started to use the [Frohlich and Apperson \(1992\)](#) diagram, but after [Kagan \(2005\)](#) I decided to try the [Kaverina et al. \(1996\)](#) one. From the original Matlab program I jumped to the Free Software world adapting it to Octave. The program only produced the x and y positions of the events and all the plotting was done by means of GMT ([Wessel et al., 2013](#)).

Some colleagues wanted to use the diagram for their work, but they were not familiar with GMT, so I decided to make a big improvement in the program to make it easy to use, distributable and with plotting support. I choose to program it in Python with the following basic ideas:

1. It should be called from the terminal or the command line in order to be incorporated into shell scripts
2. It should behave like any other shell unix tool, compatible with redirection, piping and ASCII format
3. it should be compatible with the GMT `psmeca` formats to allow the mapping of the focal mechanisms
4. It should has the option to produce a [Kaverina et al. \(1996\)](#) type classification diagram

Part of the programming was done during my happy days as Ph.D. candidate at the UCM; so I have to acknowledge the UCM scholarship that allowed me to start my scientific career.

I would like to thank the beta testers Jorge L. Giner-Robles and Alberto Jiménez-Díaz for their comments and suggestions. Dr. Andrei Bala (National Institute for Earth Physics, Romania) has tested different versions and suggested several improvements.

If you use FMC, and consider it appropriate, you can acknowledge it by citing the following reference:

- Álvarez-Gómez, J. A. (2019). FMC—Earthquake focal mechanisms data management, cluster and classification. *SoftwareX*, 9, 299-307. <https://doi.org/10.1016/j.softx.2019.03.008>

You should cite the SoftwareX paper if you are using the last version of the program with clustering and several plot customization options.

BibTeX entry of the SoftwareX paper:

```
@article{AlvarezGomez2019FMC,  
  author = {{\text{A}}lvarez-G{\text{o}}mez, Jos{\text{e}} A.},  
  title = {{FMC}---Earthquake focal mechanisms data  
    management, cluster and classification},  
  journal = {SoftwareX},  
  volume = {9},  
  pages = {299--307},  
  year = {2019},  
  doi = {10.1016/j.softx.2019.03.008}  
}
```

Part I

The manual

1 Introduction

This is the user manual for the program FMC (from Focal Mechanisms Classification). This program was originally developed on Python 2.7.3 and adapts some of the [Gasperini and Vannucci \(2003\)](#) FORTRAN routines to obtain the different parameters of the earthquake focal mechanisms depending on the input format. Since version 1.3 it is compatible with Python 3. Several output formats are eligible, clustering analysis of the data can be performed and a classification diagram for the focal mechanisms is produced; a source-type diagram of the moment tensors can also be plotted. The default input and output formats are the same used by the GMT program `psmecca` in order to make the program integration easiest and facilitate the mapping of the data.

The program basically takes the focal mechanisms data, computes the different parameters that can be obtained, classifies each focal mechanism in one of seven possible types, perform a clustering analysis of the data if it is required by the user and optionally outputs the parameters in different formats and generates a classification diagram from the input data.

2 Versions

1.10 New features:

- New parameter `Gamma`: CLVD index of [Kagan and Knopoff \(1985\)](#), computed from the deviatoric eigenvalues. It is included in the ALL output after `fclvd`, so the following columns move one position.
- The left label of the source-type diagram is now “CLVD (+)”, and the symbol of maximum magnitude of its legend keeps its position.

Changes:

- The CLVD ratio `fclvd` is computed from the deviatoric eigenvalues and follows the definition of [Frohlich and Davis \(1999\)](#): it is positive for tension-dominated CLVD, as in the source-type diagram and in [Vavryčuk \(2015\)](#). Its sign is opposite to that of versions 1.8.1 to 1.9.1 (section 8).

1.9.1 Bug fixes:

- Labels (-pa) with NumPy 2.x; source-type diagram (-pd) with a single event, equal magnitudes or PT input; no NaN for mechanisms at the centre of the classification diagram.
- -pg without a value uses its default (10); -pc with a non-numeric parameter (such as an alphanumeric ID) gives a warning and white symbols; the default plot title is the file name without path or extension.
- Clear error message when the clustering methods centroid, median or ward are used with a non-Euclidean metric.
- Magnitude legend formatted as 8.0 (previously 8.); the text label of the source-type diagram is no longer clipped; the version number shown by -h is correct.

1.9 New features:

- The symbol sizes of the diagrams are scaled to the magnitude range of the input data; the legend shows the minimum and maximum Mw.

1.8.1 Changes:

- The CLVD ratio `fcldv` follows the definition of [Giardini \(1984\)](#): it is now a signed value between -0.5 and +0.5 (previously its absolute value, as in [Frohlich and Apperson, 1992](#)). The sign convention was reversed in version 1.10.

1.8 New features:

- New output parameter `fiso`: ratio between the isotropic component and the scalar seismic moment.

1.7 New features:

- New input format PT: orientation of the P and T axes, for instance strain axes obtained from fault-slip analysis.
- New parameter `data1`: a free numeric value carried with the PT input, usable to colour or label the symbols.

1.6 New features:

- Source-type diagram of [Hudson et al. \(1989\)](#) (flag `-pd`) and new parameters `u_Hudson` and `v_Hudson`.

1.5.1 Bug fixes:

- Correction in the data reading routine (`genfromtxt`).

Changes:

- Adjustment of the T and B axes labels in the diagram.

1.5 Bug fixes:

- Several corrections.

New features:

- Warning when a non-numeric parameter is used to fill the symbols.

1.4 New features:

- Custom plot title (flag `-pt`).

Bug fixes:

- Several corrections.

1.3 New features:

- Added support for Python 3.
- Added new base diagram labels and tick marks.
- Added symbol labeling.
- Added gridlines plotting.
- Added isotropic component output.

1.2 New features:

- Added slip sense and plunge parameters for each focal mechanism nodal plane.

1.1 New features:

- New custom output parsing. The user can choose between the standard predefined output or any parameters in any order.
- Added a header with the parameter name in the output.
- Added symbol coloring.
- Added hierarchical clustering with several methods and options.

1.01 New features:

- Column number is now included in the header of the output file with all the parameters.

Bug fixes:

- Error in the `nd2ar` function to obtain the tensor components from the nodal planes.

1.0 Initial release of FMC with basic plotting function and focal mechanisms management.

3 Installation

FMC has been programmed in Python 3 and uses several common python libraries: `sys`, `argparse`, `os`, `NumPy`, `SciPy` and `matplotlib`. Depending on your operative system and Python installation you could need to install some of the libraries manually.

For **Windows** and **Mac** users a good starting point is the installation of predefined packages. From www.SciPy.org you can download program packages that includes Python, NumPy, SciPy and matplotlib, as well as other modules and IDEs. With one of these packages installed FMC should work properly.

For **Linux** users all the required software can be easily installed through your favorite distribution repository (the python modules `sys`, `argparse` and `os` are usually installed with the default python installation).

The libraries can also be installed with pip: `pip install numpy scipy matplotlib`.

FMC is formed by three files that must be kept in the same directory: `FMC.py` (main program), `functionsFMC.py` (computational functions) and `plotFMC.py` (plotting functions). The repository includes a small test file, `jap_select.dat`, with 16 large earthquakes from Japan in Global CMT format. The command `FMC.py jap_select.dat` should print the 16 events with a new column with their rupture type.

4 Usage

The user interface is as simple as a “command line”, or a “terminal”. I decided to keep the program interface as simple as possible so it can be integrated into scripts with other programs such as **GMT**, **AWK** (or **gawk**) or any other tool running from a shell or script (in *nix) or a command window or batch file (in Windows/DOS). In the following the commands are written assuming that your python installation recognizes automatically the python format. In some systems, depending on your personal configuration, you will need to call first Python (like “`python FMC.py...`” or “`python3 FMC.py...`”) or in a *nix system with the “`./`” needed to execute a script (like `./FMC.py ...`).

In *nix you will need to give execution permission to the program:

```
chmod +x FMC.py
```

To run FMC on a file simply type in the terminal:

```
FMC.py input-file.dat
```

By default FMC will read the input file as a **psmeca** Centroid Moment Tensor (CMT) format (in Harvard convention). This kind of file can be downloaded directly from **Global CMT**. The output will be shown on the screen in **psmeca** CMT format so it can be directly pipe into **psmeca**:

```
FMC.py input-file.dat | psmeca -R {...etc}
```

Or stored as an ASCII file:

```
FMC.py input-file.dat > output-file.dat
```

In this case FMC will add a new column at the end of each register with the focal mechanism type.

Alternatively the input file can be a single plane in [Aki and Richards \(1980\)](#) convention (Right Hand Rule for the plane orientation, and rakes from 0 to 180 in reverse faults and 0 to -180 in normal faults, 0 for left-lateral and ± 180 for right-lateral). This format is also compatible with `psmecca`.

4.1 Input

FMC input can be given as an ASCII file or as standard input, from a pipe ("`|`") or a redirection ("`<`"). The following codes are then equivalent:

```
FMC.py input-file.dat

cat input-file.dat | FMC.py

FMC.py < input-file.dat
```

If no input is given, then FMC will show the on-screen help.

The input format is specified with the optional flag "`-i`", and the possible values are:

CMT Harvard Centroid Moment Tensor (by default):

[longitude, latitude, depth, mrr, mtt, mff, mrt, mrf,
mtf, Exponent (dyn-cm), X plot, Y plot (for GMT), ID]

AR Aki and Richards one plane convention:

[longitude, latitude, depth, strike, dip, rake, magnitude
(Mw), X plot, Y plot (for GMT), ID]

P Both nodal planes and scalar seismic moment:

[longitude, latitude, depth, strike A, dip A, rake A,
strike B, dip B, rake B, Scalar seismic moment mantissa,
Exponent (dyn-cm), X plot, Y plot (for GMT), ID]

PT Orientation of the P and T axes:

[longitude, latitude, trend P, plunge P, trend T, plunge
T, data1, X plot, Y plot (for GMT), ID]

The data are whitespace-separated columns, one event per line. Lines starting with `#` are ignored, so the header written by FMC does not disturb a later reading. Every line must have the exact number of columns of the chosen format, and the ID must not contain spaces. With the P format the axes and the moment tensor are computed from plane A.

The PT format is intended for tensors that are not earthquake focal mechanisms, such as the shortening (P) and extension (T) axes obtained from fault-slip inversion; the P and T axes must be orthogonal. `data1` is any numeric value (a site number, the number of faults, a misfit...) that can be used to colour (`-pc data1`) or label (`-pa data1`) the symbols or to perform the clustering (`-ci data1`). As there is no depth or magnitude information FMC assigns depth 0, $M_0 = 10^{20}$ dyn·cm and Mw 8.0.

4.2 Output

FMC output format can be selected among the following options with the flag “-o”:

CMT Harvard Centroid Moment Tensor (psmecca compatible):

[longitude, latitude, depth, mrr, mtt, mff, mrt, mrf, mtf, Exponent (dyn·cm), X plot, Y plot (for GMT), ID, TYPE]

P Focal mechanism both nodal planes (psmecca compatible):

[longitude, latitude, depth, strike A, dip A, rake A, strike B, dip B, rake B, Scalar seismic moment mantissa, Exponent (dyn·cm), X plot, Y plot (for GMT), ID, TYPE]

AR Focal mechanism one plane (psmecca compatible):

[longitude, latitude, depth, strike, dip, rake, magnitude (Mw), X plot, Y plot (for GMT), ID, TYPE]

K Kaverina diagram position for plotting outside FMC:

[X Kaverina diagram, Y Kaverina diagram, Mw, Depth, ID, TYPE]

ALL All the parameters obtained:

[longitude, latitude, depth, mrr, mtt, mff, mrt, mrf, mtf, Exponent (dyn·cm), Scalar seismic moment (dyn·cm), Mw, strike A, dip A, rake A, strike B, dip B, rake B, Slip trend A, Slip plunge A, Slip trend B, Slip plunge B, P trend, P plunge, B trend, B plunge, T trend, T plunge, fclvd, Gamma, Isotropic component, Isotropic ratio, u Hudson, v Hudson, X Kaverina diagram, Y Kaverina diagram, ID, TYPE]

CUSTOM In case you need any focal mechanism parameters in any order you can use the CUSTOM option and give the requested parameters in any order using the flag “-of”. The output parameters need to be listed separated by commas. The accepted parameters

names are listed below, and can be seen on the terminal using `FMC.py -helpFields`

lon longitude

lat latitude

dep depth

mrr mrr centroid moment tensor component

mtt mtt centroid moment tensor component

mff mff centroid moment tensor component

mrt mrt centroid moment tensor component

mrf mrf centroid moment tensor component

mtf mtf centroid moment tensor component

mant mantissa of the seismic moment tensor

expo exponent of the seismic moment tensor

Mo Scalar seismic moment

Mw Moment (or Kanamori) magnitude

strA Strike of nodal plane A

dipA Dip of nodal plane A

rakeA Rake of nodal plane A

strB Strike of nodal plane B

dipB Dip of nodal plane B

rakeB Rake of nodal plane B

slipA Trend of the slip vector of plane A

plungA Plunge of slip vector of plane A

slipB Trend of the slip vector of plane B

plungB Plunge of slip vector of plane B

trendp Trend of P axis

plungp Plunge of P axis

trendb Trend of B axis

plungb Plunge of B axis

trendt Trend of T axis

plungt Plunge of T axis

fclvd Compensated linear vector dipole ratio (positive for tension;
section 8)

Gamma CLVD index of [Kagan and Knopoff \(1985\)](#)

iso Isotropic component of the Moment Tensor
fiso Ratio between the isotropic component and the scalar seismic moment
u_Hudson u position on the source-type diagram
v_Hudson v position on the source-type diagram
x_kav x position on the Kaverina diagram
y_kav y position on the Kaverina diagram
ID ID of the event
clas Focal mechanism rupture type
posX X plotting position for GMT psmeca
posY Y plotting position for GMT psmeca
clustID Cluster number (0 if no clustering analysis is done)
data1 Free numeric value of the PT input (0 otherwise)

All the output formats start with a header line, beginning with #, with the name of each column. Except CUSTOM, all of them add the rupture type (TYPE) as last column, and the cluster number (Cluster_ID) when a clustering analysis is done. In the CMT output the tensor components are rescaled to the exponent of the computed scalar moment, so the output exponent may differ from the input one.

Columns of the ALL output (useful to select fields with awk):

#	Header	#	Header	#	Header
1	Longitude	14	Dip_A	27	Trend_T
2	Latitude	15	Rake_A	28	Plunge_T
3	Depth_(km)	16	Strike_B	29	fclvd
4	mrr	17	Dip_B	30	Gamma
5	mtt	18	Rake_B	31	Isotropic
6	mff	19	Slip_trend_A	32	Iso_ratio
7	mrt	20	Slip_plunge_A	33	u_Hudson
8	mrf	21	Slip_trend_B	34	v_Hudson
9	mtf	22	Slip_plunge_B	35	X_Kaverina
10	Exponent_(dyn-cm)	23	Trend_P	36	Y_Kaverina
11	Seismic_moment_Mo	24	Plunge_P	37	ID
12	Magnitude_Mw	25	Trend_B	38	rupture_type
13	Strike_A	26	Plunge_B	39	Cluster_ID (if clustering)

Examples of use

Obtaining nodal planes from moment tensor

Command:

```
echo -2.54 37.09 12 -3.4669 -2.0652 5.5321 6.2368
      -1.8004 -5.1775 22 X Y ID | FMC.py -o P
```

Result:

```
#Longitude Latitude Depth_(km) Strike_A Dip_A Rake_A
  Strike_B Dip_B Rake_B Seismic_moment_mantissa
  Exponent_(dyn-cm) X_position(GMT) Y_position(GMT)
  ID rupture_type
-2.54 37.09 12.0 190.925 42.4899 -20.9735 296.709
  76.0089 -130.541 9.611967 22.0 X Y ID N-SS
```

Obtaining moment tensor from one nodal plane

Command:

```
echo -2.54 37.09 12 190.925 42.4899 -20.9735 4.6 X Y ID
      | FMC.py -i AR -o CMT
```

Result:

```
#Longitude Latitude Depth_(km) mrr mtt mff mrt mrf mtf
  Exponent_(dyn-cm) X_position(GMT) Y_position(GMT)
  ID rupture_type
-2.54 37.09 12.0 -3.56563 -2.21928 5.78491 6.70126
  -1.61249 -5.19047 22.0 X Y ID N-SS
```

Obtaining all the parameters from the moment tensor

Command:

```
echo -2.54 37.09 12 -3.4669 -2.0652 5.5321 6.2368
      -1.8004 -5.1775 22 X Y ID | FMC.py -o ALL
```

Result:

```
#Longitude Latitude Depth_(km) mrr mtt mff mrt mrf mtf
  Exponent_(dyn-cm) Seismic_moment_Mo Magnitude_Mw
  Strike_A Dip_A Rake_A Strike_B Dip_B Rake_B
  Slip_trend_A Slip_plunge_A Slip_trend_B
  Slip_plunge_B Trend_P Plunge_P Trend_B Plunge_B
  Trend_T Plunge_T fclvd Gamma Isotropic Iso_ratio
  u_Hudson v_Hudson X_Kaverina Y_Kaverina ID
  rupture_type
-2.54 37.09 12.0 -3.4669 -2.0652 5.5321 6.2368 -1.8004
  -5.1775 22.0 9.61197e+22 4.6 190.925 42.4899
  -20.9735 296.709 76.0089 -130.541 206.709 -13.9911
  100.925 -47.5101 167.141 43.8185 308.393 39.1024
```

```
56.0979 20.5155 0.0445259 0.117979 -1398100.0
-1.45454e-17 -0.0890517 0.0 -0.243839 0.0899979 ID
N-SS
```

Obtaining all the parameters from CMT input file and storing to an ASCII file

Command:

```
FMC.py -o ALL jap_select.dat > Japan_parameters.dat
```

Using CUSTOM output to obtain event location and slip vector of both nodal planes

Command:

```
echo -2.54 37.09 12 -3.4669 -2.0652 5.5321 6.2368
      -1.8004 -5.1775 22 X Y ID | FMC.py -o CUSTOM -of
lon,lat,slipA,plungA,slipB,plungB
```

Result:

```
#Longitude Latitude Slip_trend_A Slip_plunge_A
      Slip_trend_B Slip_plunge_B
-2.54 37.09 206.709 -13.9911 100.925 -47.5101
```

Obtaining the CLVD parameters of a catalogue

Command:

```
FMC.py -o CUSTOM -of ID,Mw,fclvd,Gamma,clas jap_select.
dat
```

Result (first events):

```
#ID Magnitude_Mw fclvd Gamma rupture_type
032378C 7.6 0.00882022 0.0230146 R
052683A 7.7 0.122769 0.33196 R
110189E 7.4 0.050104 0.133035 R
071293B 7.7 0.0604748 0.161154 R
```

Obtaining nodal planes and B axis from P and T axes (PT input)

Command:

```
FMC.py -i PT -o CUSTOM -of ID,trendb,plungb,strA,dipA,
rakeA,strB,dipB,rakeB,clas,data1 pt_example.dat
```

Result:

```
#ID Trend_B Plunge_B Strike_A Dip_A Rake_A Strike_B
Dip_B Rake_B rupture_type data1
S1 350.0 80.0 214.561 82.947 -7.10708 305.439 82.947
-172.893 SS 12.0
S2 200.0 85.0 155.109 86.4667 -176.46 64.8908 86.4667
-3.54002 SS 25.0
S3 315.0 1.59028e-15 315.0 25.0 -90.0 135.0 65.0 -90.0
N 8.0
S4 210.0 7.95139e-15 210.0 15.0 90.0 30.0 75.0 90.0 R
40.0
S5 120.0 50.0 337.454 62.966 -30.6821 82.5463 62.966
-149.318 SS-N 30.0
```

Using an FMC output as input

The extra column with the rupture type must be removed first, for instance with awk:

```
FMC.py -o AR jap_select.dat | awk '{$NF=""; print}' |
FMC.py -i AR -o CMT
```

4.3 Plot

Optionally FMC will produce a classification diagram. FMC uses matplotlib libraries and can generate figures in different formats (emf, eps, jpeg, jpg, pdf, png, ps, raw, rgba, svg, svgz, tif, tiff). The format is determined automatically from the plot file name extension.

The diagram uses the [Kaverina et al. \(1996\)](#) projection technique, used also by [Kagan \(2005\)](#); but incorporating a classification similar to the geological conceptual classification of faults. The earthquakes are classified into seven types according to the values of the P, T and B Centroid Moment Tensor axes following a simple algorithm (Figure 1), and are represented conveniently on the Kaverina diagram (Figure 2). This classification is very similar to the used by [Johnston et al. \(1994\)](#).

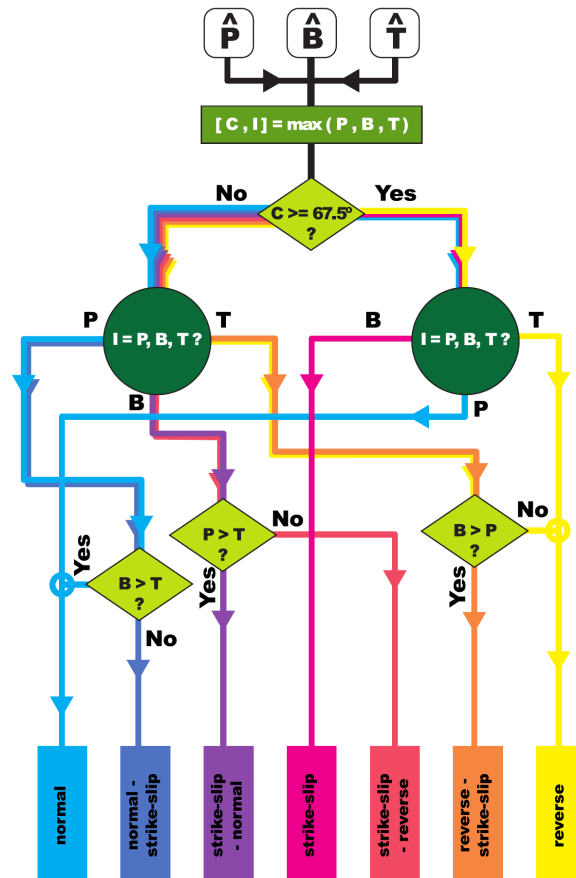


Figure 1: Focal mechanism classification algorithm flow chart.

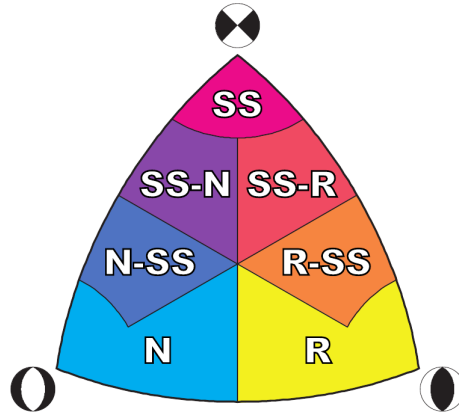


Figure 2: Classification diagram. N: Normal; N-SS: Normal - Strike-slip; SS-N: Strike-slip - Normal; SS: Strike-slip; SS-R: Strike-slip - Reverse; R-SS: Reverse - Strike-slip; R: Reverse.

The size of the symbols is proportional to the magnitude, scaled between the minimum and maximum M_w of the data, which are shown in the legend. FMC uses several flags in order to customize the plot.

- p This flag activates the plotting. It must be followed by the name of the figure file that will be produced. The name used for the file (without the extension) is used as title for the plot, unless a different one is given with -pt.
- pd This flag plots the source-type diagram of [Hudson et al. \(1989\)](#) (section 11) in the given file. The flags -pc, -pa and -pt also work with this diagram.
- pc With this flag the user specifies the parameter that is used to fill the symbols. A color palette is produced with the range of the selected parameter values. The parameter must be numeric; otherwise FMC gives a warning and draws white symbols.
- pg This flag is used to plot gridlines with the specified grid spacing (10 degrees if no value is given).
- pa This flag is used to annotate the symbols with a certain parameter.
- pt Plot title. -pt " " removes the title.

With -pc and -pa the parameters must be given with its corresponding internal name as listed in section 4.2.

Examples of use

Plotting data from standard input

Command:

```
echo -2.54 37.09 12 -3.4669 -2.0652 5.5321 6.2368  
-1.8004 -5.1775 22 X Y ID | FMC.py -p "My data.png"
```

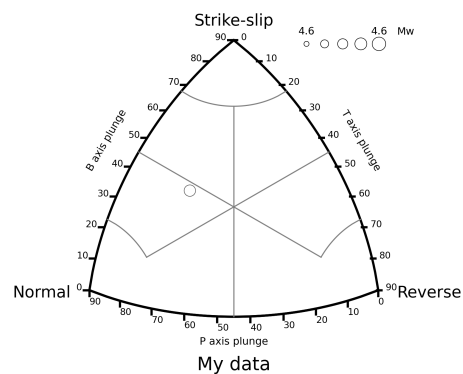


Figure 3: Plot result from the command `echo -2.54 37.09 12 -3.4669 -2.0652 5.5321 6.2368 -1.8004 -5.1775 22 X Y ID | FMC.py -p "My data.png"`

Plotting data from input file, shading the symbols with a parameter and plotting gridlines

Command:

```
FMC.py -p 'Japan data.png' jap_select.dat -pc dep -pg  
10 -pt "Japan, Mw >= 7.2"
```

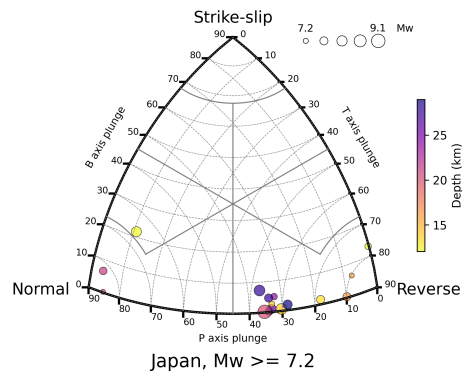


Figure 4: Plot result from the command `FMC.py -p 'Japan data.png' jap_select.dat -pc dep -pg 10 -pt "Japan, Mw >= 7.2"`

Plotting data from input file shading and annotating the symbols

Command:

```
FMC.py jap_select.dat -p 'Japan labels.png' -pc dep -pa Mw -pt "Japan, Mw >= 7.2"
```

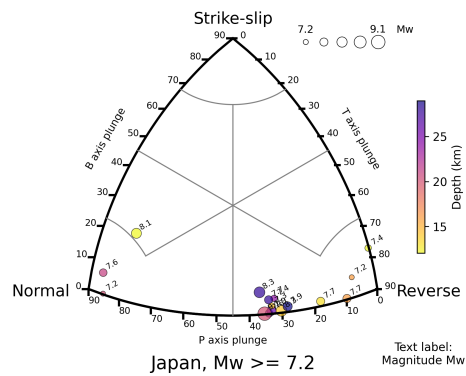


Figure 5: Plot result from the command `FMC.py jap_select.dat -p 'Japan labels.png' -pc dep -pa Mw -pt "Japan, Mw >= 7.2"`

Plotting the source-type diagram

Command:

```
FMC.py jap_select.dat -pd 'Japan source type.png' -pc fclvd -pt "Japan, source type"
```

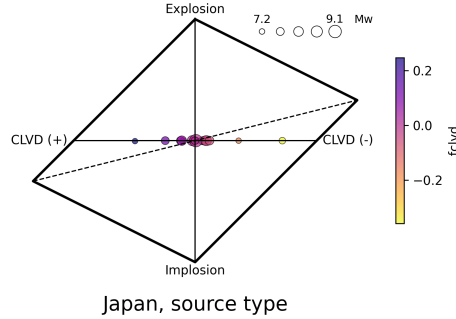


Figure 6: Plot result from the command `FMC.py jap_select.dat -pd 'Japan source type.png' -pc fclvd -pt "Japan, source type"`. The Global CMT solutions are deviatoric, so all the events lie on the horizontal axis. Symbols are coloured by `fclvd`: positive values (purple, tension-dominated CLVD) plot on the left and negative values (yellow) on the right (section 8).

Plotting P and T axes from fault-slip analysis

Command:

```
FMC.py -i PT pt_example.dat -p pt_axes.png -pc data1 -
pt "Strain axes from fault-slip data"
```

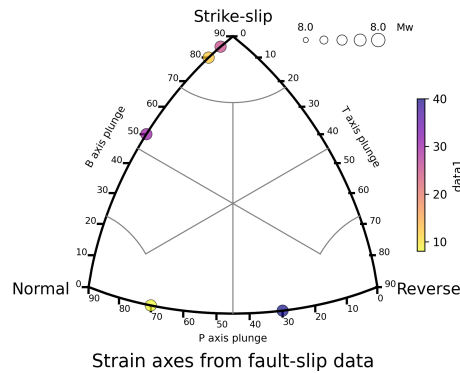


Figure 7: Plot result from the command `FMC.py -i PT pt_example.dat -p pt_axes.png -pc data1 -pt "Strain axes from fault-slip data"`. All symbols have the same size because this format has no magnitude information.

Plotting data from input file and piping to psmeca

Command:

```
FMC.py jap_select.dat | gmt psmeca -R125/170/30/60 -  
JM8c -Ba10f2WESN -Sm0.4c+f0 -Gred > CMT_map.ps
```

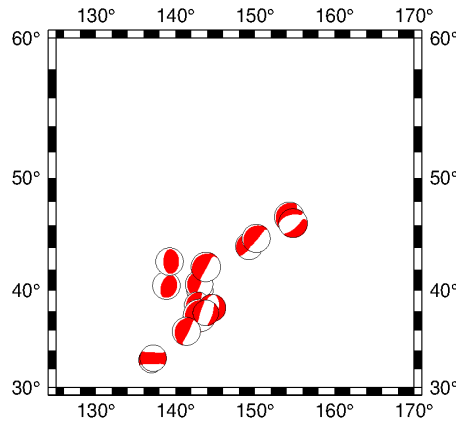


Figure 8: Map generated with psmeca (GMT) from the FMC output.

4.4 Clustering

FMC implements the hierarchical agglomerative clustering algorithms from SciPy (`scipy.cluster.hierarchy`) in order to group the data¹. The advantages of this algorithms are its versatility, as the user can choose between a number of metrics and grouping methods, its capacity to automatically select a minimum number of clusters without an a priori estimation, and its potential to work with different parameters with different scales and with strong different populations in clusters.

The parameters for the clustering are passed by several optional flags. If any of the following flags is given in the command line FMC will perform the clustering analysis using some default options if needed. When a clustering analysis is done, by default FMC will shade the symbols in the diagram using the cluster number, unless a different parameter is stated with `-pc` flag. The cluster number of each event is added as a last column (`Cluster_ID`) to the output.

`-cm` Method to be used in the clustering analysis. The options are:

¹The details on the clustering algorithms shown in this section are taken from the [SciPy documentation](#).

single single/min/nearest

$$d(u, v) = \min(\text{dist}(u[i], v[j]))$$

complete complete/max/farthest point

$$d(u, v) = \max(\text{dist}(u[i], v[j]))$$

average average/UPGMA

$$d(u, v) = \sum_{ij} \frac{d(u[i], v[j])}{(|u| * |v|)}$$

weighted weighted/WPGMA

$$d(u, v) = (\text{dist}(s, v) + \text{dist}(t, v)) / 2$$

centroid centroid/UPGMC [default]

$$\text{dist}(s, t) = \|c_s - c_t\|_2$$

where c_s and c_t are the centroids of clusters s and t , respectively. When two clusters s and t are combined into a new cluster u , the new centroid is computed over all the original objects in clusters s and t . The distance then becomes the Euclidean distance between the centroid of u and the centroid of a remaining cluster v in the forest.

median median/WPGMC, assigns $d(s, t)$ like the centroid method.

When two clusters s and t are combined into a new cluster u , the average of centroids s and t give the new centroid.

ward Ward variance minimization algorithm.

$$d(u, v) = \sqrt{\frac{|v| + |s|}{T} d(v, s)^2 + \frac{|v| + |t|}{T} d(v, t)^2 - \frac{|v|}{T} d(s, t)^2}$$

where u is the newly joined cluster consisting of clusters s and t , v is an unused cluster in the forest, $T = |v| + |s| + |t|$, and $|*|$ is the cardinality of its argument.

Methods “centroid”, “median” and “ward” are correctly defined only if Euclidean pairwise metric is used. Since centroid is the default method, any other metric given with `-ce` must be combined with `-cm` single, complete, average or weighted; otherwise FMC stops with an error message.

-**ce** Metric used to measure distances between events parameters. These metrics work with non-Boolean vectors. By default FMC uses euclidean distance.

braycurtis The Bray-Curtis distance between two points u and v is

$$d(u, v) = \frac{\sum_i |u_i - v_i|}{\sum_i |u_i + v_i|}$$

canberra The Canberra distance between two points u and v is

$$d(u, v) = \sum_i \frac{|u_i - v_i|}{|u_i| + |v_i|}$$

chebyshev The Chebyshev distance between two n-vectors u and v is the maximum norm-1 distance between their respective elements. More precisely, the distance is given by

$$d(u, v) = \max_i |u_i - v_i|$$

cityblock City block or Manhattan distance between the points.

correlation Correlation distance between vectors u and v . This is

$$1 - \frac{(u - \bar{u}) \cdot (v - \bar{v})}{\|(u - \bar{u})\|_2 \|(v - \bar{v})\|_2}$$

cosine Cosine distance between vectors u and v ,

$$1 - \frac{u \cdot v}{\|u\|_2 \|v\|_2}$$

euclidean Distance between m points using Euclidean distance (2-norm). [Default]

hamming Normalized Hamming distance, or the proportion of those vector elements between two n-vectors u and v which disagree.

jaccard Jaccard distance between the points. Given two vectors, u and v , the Jaccard distance is the proportion of those elements $u[i]$ and $v[i]$ that disagree.

mahalanobis The Mahalanobis distance between two points u and v is

$$\sqrt{(u - v)(1/V)(u - v)^T}$$

where $1/V$ is the inverse covariance matrix.

minkowski Distances using the Minkowski distance $\|u - v\|_p$ (p-norm) where $p \geq 1$.

seuclidean Standardized Euclidean distance. The standardized Euclidean distance between two n-vectors u and v is

$$\sqrt{\sum (u_i - v_i)^2 \div V[x_i]}$$

V is the variance vector; $V[i]$ is the variance computed over all the i 'th components of the points. It is automatically computed.

squclidean Squared Euclidean distance $\|u - v\|_2^2$ between the vectors.

- cn A priori number of clusters to obtain. If zero or non-present the number of clusters is automatically computed with an elbow criterion on the dendrogram: the merging distances are taken from the last merge to the first and the number of clusters is set where their second difference is largest (the minimum is two clusters).
- ci Parameters used to perform the cluster analysis. By default FMC uses the position on the Kaverina diagram, which is a proxy for the principal moment tensor axes plunges. If the parameters given are not in the same physical magnitude and unit the euclidean distance is not appropriate and a different metric should be used. In these cases the Mahalanobis distance is a good choice, as is equivalent to the Euclidean distance in the transformed space, using the covariance matrix of each parameter. Angular parameters such as strikes and trends are circular (0° and 360° are the same direction), which is not taken into account by any of these metrics.

The parameters must be given with its corresponding internal name as listed in section 4.2.

Examples of use

Automatic clustering using the position in the Kaverina diagram (default)

Command:

```
FMC.py -p 'Japan clusters.png' jap_select.dat -cn 0 -pt  
"Japan, automatic clustering"
```

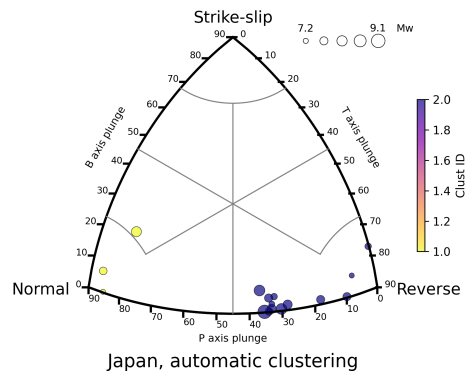


Figure 9: Plot result from the command `FMC.py -p 'Japan clusters.png' jap_select.dat -cn 0 -pt "Japan, automatic clustering"`

The output contains the cluster of each event in the last column:

```
#ID Magnitude_Mw rupture_type Cluster_ID
032378C 7.6 R 2
052683A 7.7 R 2
110189E 7.4 R 2
071293B 7.7 R 2
```

Clustering using the epicentral location

Command:

```
FMC.py jap_select.dat -cn 3 -ci lon,lat > clusters.dat
```

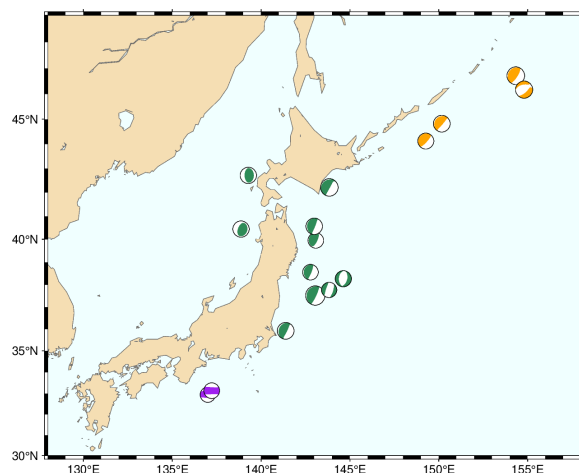


Figure 10: Plotting with GMT of the clusters obtained with the command `FMC.py jap_select.dat -cn 3 -ci lon,lat` (see section 4.5).

Clustering using Mahalanobis metric and complete grouping with the slip and plunge of the slip vector of nodal plane A

Command:

```
FMC.py -p 'Japan clusters plane A.png' jap_select.dat -
cn 3 -ci slipA,plunga -cm complete -ce mahalanobis
```

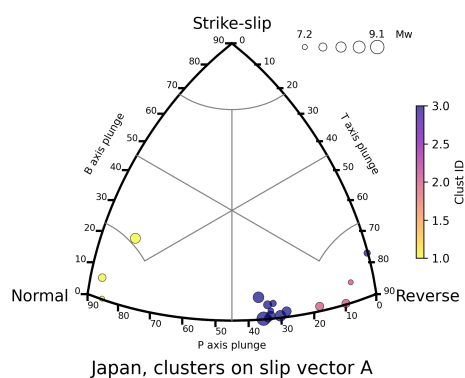


Figure 11: Plot result from the command `FMC.py -p 'Japan clusters plane A.png' jap_select.dat -cn 3 -ci slipA,plunga -cm complete -ce mahalanobis`

4.5 Working with GMT

The CMT, P and AR outputs can be read directly by the GMT module `meca` (`psmeca` in GMT 4 and 5), with the symbol options `-Sm`, `-Sc` and `-Sa` respectively. The header line starts with `#` and is skipped by GMT. The last columns (ID, rupture_type and, if present, Cluster_ID) are read as the event label; add `+f0` to the symbol option to hide the labels.

```
FMC.py jap_select.dat | gmt psmeca -R125/170/30/60 -  
    JM8c -Ba10f2WESN -Sm0.4c > CMT_map.ps  
FMC.py jap_select.dat | gmt meca -R128/158/30/49 -JM15c  
    -Baf -Sm0.3c -png japan_map
```

The rupture type is written in column 14 of the CMT output, so the mechanisms can be coloured by tectonic regime with `awk`. Here `/^N/` selects the classes N and N-SS, `/^SS/` selects SS, SS-N and SS-R, and `/^R/` selects R and R-SS:

```
gmt begin japan_rupture_type png  
gmt coast -R128/158/30/49 -JM15c -Baf -BWSne -Gwheat  
    -Sazure -W0.25p,gray40  
FMC.py jap_select.dat | awk '$14 ~ /^N/' | gmt meca  
    -Sm0.3c+f0 -Gdodgerblue -W0.3p  
FMC.py jap_select.dat | awk '$14 ~ /^SS/' | gmt meca  
    -Sm0.3c+f0 -Gforestgreen -W0.3p  
FMC.py jap_select.dat | awk '$14 ~ /^R/' | gmt meca  
    -Sm0.3c+f0 -Gred -W0.3p  
gmt end show
```

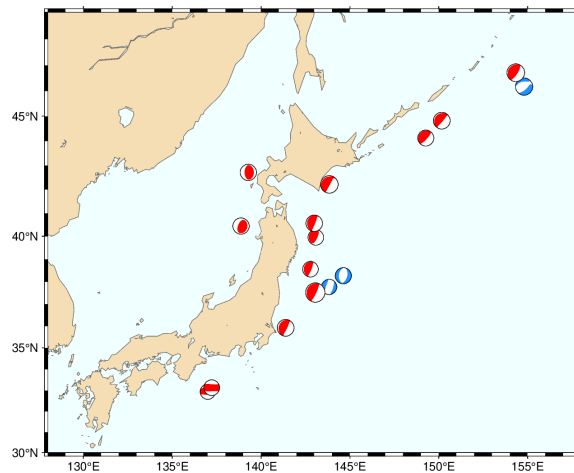


Figure 12: GMT map of the test data coloured by regime: normal and normal – strike-slip in blue, reverse in red.

Similarly, the clusters can be mapped using the `Cluster_ID` column (column 15 of the CMT output):

```
FMC.py jap_select.dat -cn 3 -ci lon,lat > clusters.dat
gmt begin japan_spatial_clusters png
gmt coast -R128/158/30/49 -JM15c -Baf -BWSne -Gwheat
  -Sazure -W0.25p,gray40
awk '$15==1' clusters.dat | gmt meca -Sm0.3c+f0 -
  Gorange -W0.3p
awk '$15==2' clusters.dat | gmt meca -Sm0.3c+f0 -
  Gpurple -W0.3p
awk '$15==3' clusters.dat | gmt meca -Sm0.3c+f0 -
  Gseagreen -W0.3p
gmt end show
```

4.6 Command-line reference

```
FMC.py [infile] [-i {CMT,AR,P,PT}] [-o {CMT,P,AR,K,ALL,
CUSTOM}] [-of FIELDS]
  [-p PLOTFILE] [-pd PLOTFILE] [-pc PARAM] [-pa
PARAM] [-pg SPACING] [-pt TITLE]
  [-cm METHOD] [-ce METRIC] [-cn N] [-ci FIELDS]
  [-v] [-helpFields] [-h]
```

Flag	Description	Section
<code>infile</code>	Input file; if absent, FMC reads the standard input	4.1
<code>-i</code>	Input format: CMT (default), AR, P, PT	4.1
<code>-o</code>	Output format: CMT (default), P, AR, K, ALL, CUSTOM	4.2
<code>-of</code>	Comma-separated parameters for <code>-o</code> CUSTOM	4.2
<code>-p</code>	Classification diagram file	4.3
<code>-pd</code>	Source-type diagram file	4.3
<code>-pc</code>	Parameter used to colour the symbols	4.3
<code>-pa</code>	Parameter used to label the symbols	4.3
<code>-pg</code>	Gridline spacing in degrees	4.3
<code>-pt</code>	Plot title	4.3
<code>-cm</code>	Clustering linkage method	4.4
<code>-ce</code>	Clustering distance metric	4.4
<code>-cn</code>	Number of clusters (0: automatic)	4.4
<code>-ci</code>	Comma-separated clustering parameters	4.4
<code>-v</code>	Verbose: progress information on the standard error	—
<code>-helpFields</code>	Lists the parameter names and exits	4.2
<code>-h</code>	Shows the help and exits	—

4.7 Troubleshooting and known issues

“ERROR - Incorrect number of columns”. Each input format requires an exact number of columns (13, 10, 14 or 10 for CMT, AR, P and PT). Check the chosen `-i` format, remove extra columns (such as the `rupture_type` column of a previous FMC output) and make sure that event IDs contain no spaces.

“ERROR - The clustering methods ‘centroid’, ‘median’ and ‘ward’ only work with the ‘euclidean’ metric”. A non-Euclidean metric was chosen with `-ce` while keeping the default linkage method. Add `-cm single`, `complete`, `average` or `weighted`, or use `-ce euclidean`.

“Warning, to fill the symbols a numeric value is needed”. The parameter chosen with `-pc` is not numeric (for example `clas` or alphanumeric event IDs); the symbols are drawn in white.

BrokenPipeError when piping into head. Harmless: the output was interrupted by the receiving program after it read what it needed.

PT input gives unexpected planes. The P and T axes must be orthogonal; FMC does not check or correct this.

Errors in versions before 1.9.1. In FMC 1.9 and earlier, `-pa` fails with `TypeError: only 0-dimensional arrays can be converted to Python scalars` when using NumPy 2.x, and `-pd` fails with a single event or when all the magnitudes are equal (always with PT input). Update to version 1.9.1 or later (section 2).

5 License

FMC has been developed with Free Software and/or Open Source tools. FMC uses NumPy, SciPy and matplotlib which are distributed under BSD license.

FMC is distributed under [GNU General Public License v3.0](#). This manual is distributed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International license ([CC BY-NC-SA 4.0](#)).

Part II

The background

6 Introduction

Seismicity is frequently used to deduce the tectonics of a region. The study of earthquakes as a tectonic component, the seismotectonics (Scholz, 2002), has grown as one of the key research areas on active tectonics, specially from the analysis of earthquake focal mechanisms.

The first focal mechanism determination methods, based on P wave's first motion polarities, were developed on the first half of the 20th century, specially in Japan where a dense seismic network were available (Byerly, 1928, 1938; Koning, 1941; Honda and Masatuke, 1952; Scheidegger, 1957). Since 1960 the development of the digital computer allow the numerical determination of fault-plane solutions with different methods (e.g. Knopoff, 1961; Langston and Helmberger, 1975; Dziewonski et al., 1981).

In parallel with the development of the modern seismology, the theory of Plate Tectonics changed the way geologists understand the Earth. Since the end of the 16th century naturalists made observations on the potential continental drifting; but it was in the early 20th century when the theory started to be formally proposed (Taylor, 1910; Wegener, 1912; Schwinner, 1919; Holmes, 1931). The relevant information acquired on the magnetic stripes on the ocean floor in the 60's (e.g. Hess, 1962; Vine, 1966; Le Pichon, 1968; Morgan, 1968), in addition to an increasing number of geological and geophysical observations, established the Plate Tectonics as the global geodynamical paradigm on Earth Sciences. The study of earthquakes related to Plate Tectonics was developed at the same time establishing the basic concepts of seismotectonics and the lithospheric deformation (e.g. Benioff, 1954; Sykes, 1967; Isacks et al., 1968; Isacks and Molnar, 1969).

Since the 70's focal mechanisms are computed in a systematic way and global catalogs of focal mechanisms are available since then. As the amount of data increases, our knowledge on the active tectonics improves. As a consequence of the continuous increase of data available we need new tools to analyze it systematically.

In order to represent focal mechanism populations Frohlich and Apperson (1992) proposed a diagram to visualize focal mechanism data as a function of the rupture type. This kind of representation is popular and widely used on seismotectonics to represent the focal mechanism types of the study areas (e.g. Frohlich et al., 1997; Borges et al., 2001; Igarashi et al., 2001; Ratchkovski and Hansen, 2002; Kita et al., 2006; Serpelloni et al., 2007). This

representation is not accurate, presenting distortion problems (Frohlich, 2001), and Kagan (2005) used the Kaverina et al. (1996) projection to avoid them. The latter projection is used in FMC by default.

7 The focal mechanism

The term “focal mechanism” is used to refer to the parameters that characterize an earthquake rupture. A focal mechanism usually presents the characteristics of the two orthogonal possible rupture planes: strike, dip and rake of the slip vector over the plane. The energy released by the event, its hypocentral location and the exact time of occurrence are also given. The focal mechanism can, instead of describe the geometry of the possible rupture planes (6 variables), describe the characteristics of the rupture by means of the seismic moment tensor (SMT) or centroid moment tensor (CMT) (6 variables).

The SMT is based in the representation of the equivalent forces on a point seismic source. In this tensor the sum of angular moments should be null, in other words, the SMT is a symmetric tensor on a three dimensional space (9 components) with 6 independent components:

$$\mathbf{M} = \begin{pmatrix} M_{XX} & M_{XY} & M_{XZ} \\ M_{YX} & M_{YY} & M_{YZ} \\ M_{ZX} & M_{ZY} & M_{ZZ} \end{pmatrix} \quad (7.1)$$

where

$$M_{XY} = M_{YX}, \quad M_{XZ} = M_{ZX}, \quad M_{YZ} = M_{ZY} \quad (7.2)$$

Similarly to the strain tensor where we can define the orientation of the principal axes (eigenvectors) and its magnitudes (eigenvalues), the SMT can be defined by the orientation of its principal axes P, B and T (eigenvectors) and its magnitudes (eigenvalues). The T axis describes the greatest of the eigenvalues, the P describes the lowest and the B the intermediate.

If the seismic event is generated by the slip in a fault rupture, its characteristics should be well described as a double couple moment tensor (Figure 13):

$$\mathbf{M} = \begin{bmatrix} M_0 & 0 & 0 \\ 0 & -M_0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (7.3)$$

where M_0 is the scalar seismic moment.

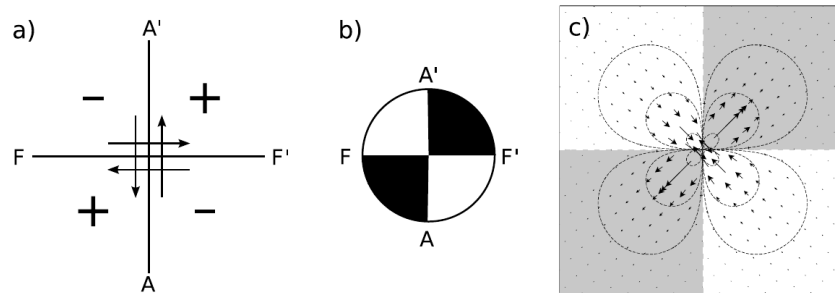


Figure 13: Double couple deformation mechanism. a) Plan view of horizontal displacement on an idealized point vertical fault A-A' or F-F' and resulting distribution of compressions (+) and dilatations (-). b) Focal mechanism representation on a stereographic projection. Compression quadrants are filled. c) Horizontal displacement produced by a point size vertical dislocation computed with the [Okada \(1992\)](#) equations; compressional quadrants are filled while dilatational are kept empty following the focal mechanism representation. a) and b) modified from [Bullen and Bolt \(1985\)](#).

The most used representation of the double couple is based on the stereographic projection. In this projection both nodal planes of the focal mechanism are plotted, and the quadrants containing the T axis, the compression quadrants, are filled (Figure 13). This kind of representation - commonly called "beach-ball" - located on a map on the earthquake epicenter is useful to interpret the seismic data from a tectonic point of view, allowing the study of the earthquake and its rupture fault plane on its tectonic context, being one of the basis of the seismotectonics. The compressional and dilatational quadrants are separated by two orthogonal planes, the nodal planes, being one of them the plane of rupture that generated the earthquake (Figure 14).

As we increase the number of focal mechanisms to study, we increase the difficulty to interpret the data, obscuring frequently the details of a tectonic environment if we just project them over a map. In order to facilitate the study of groups of focal mechanisms several plots can be proposed, great part of them based on classical structural geology plots.

The plots inherited from the structural geology are based in the geometrical characteristics of the nodal planes of the SMT, equivalent to the characteristics of the fault planes: strike (φ), dip (δ) and rake (λ) (Figure 14). The main problem of these representations is the duplication of data. Each focal mechanism presents two equivalent and orthogonal nodal planes. The selection of one of them as the fault plane is not straightforward and although from a mechanical point of view they can be discriminated ([McKenzie, 1969](#); [Michael, 1987](#); [Gephart, 1990](#)), when both nodal planes are me-

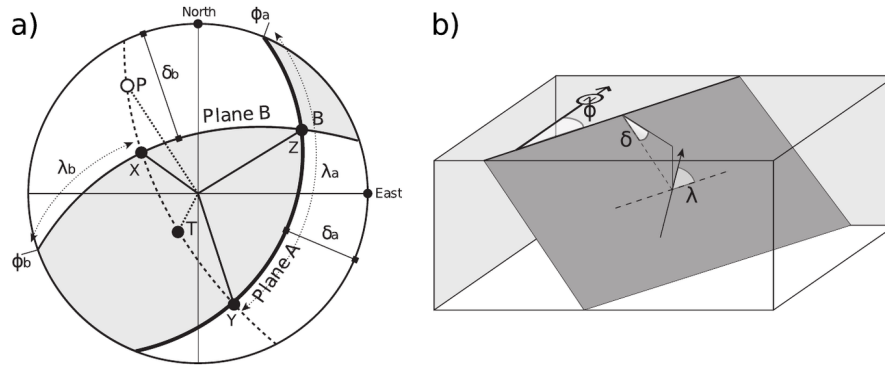


Figure 14: a) Focal sphere diagram on stereographic projection on the lower hemisphere. Nodal planes A and B, A representing the fault plane and B the auxiliary plane, with its orientation characteristics: ϕ , strike; δ , dip and λ , rake. The orientation of the SMT principal axes are also shown: T, B and P. Modified from Udías (1989). b) Block diagram showing the parameters that define the orientation of a fault plane in the space: ϕ , strike; δ , dip and λ , rake.

chanically compatible or reactivated structures can be present, additional geological and/or geophysical information is required. There are as many representations of this kind as possible combinations between these parameters and its derivatives (usually trigonometrical functions). These relations can be represented in biaxial plots or stereographic projections (e.g. Davis and Reynolds, 1996; Ramsay and Huber, 1997; Pollard and Fletcher, 2005).

The focal mechanisms can also be plotted as function of the principal axes T, B and P. These axes form an orthogonal system. The orientation of each axis can be defined in the space as any line, by its direction and plunge. The main advantages of these representations are: the univocality, each focal mechanism is represented by one point; its simplicity of interpretation, the kind of deformation (normal, reverse, strike-slip) is function of the plunge of the axes; and the reduction of errors compared with the planes, as they are derived from the principal axes of the SMT (Vannucci and Gasperini, 2003).

8 Parameters computed by FMC

This section describes how FMC obtains each of the parameters listed in section 4.2.

Moment tensor. FMC works internally in the Aki and Richards (1980)

Cartesian system (x north, y east, z down). The Harvard components (r up, θ south, ϕ east) are converted as:

$$\mathbf{M} = \begin{pmatrix} M_{\theta\theta} & -M_{\theta\phi} & M_{r\theta} \\ -M_{\theta\phi} & M_{\phi\phi} & -M_{r\phi} \\ M_{r\theta} & -M_{r\phi} & M_{rr} \end{pmatrix}$$

With AR, P and PT input, a double-couple tensor is built from the unit normal $\mathbf{n}$ and slip $\mathbf{d}$ vectors of nodal plane A as $M_{ij} = M_0 (d_i n_j + d_j n_i)$.

Principal axes. The eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3$ of $\mathbf{M}$ and their eigenvectors define the P, B and T axes respectively. The trend (azimuth, 0–360°) and plunge (0–90°, downwards) of each axis are given.

Nodal planes and slip vectors. The unit normal and slip vectors of the nodal planes are obtained from the P and T unit vectors as $\mathbf{n} \propto \mathbf{T} + \mathbf{P}$ and $\mathbf{d} \propto \mathbf{T} - \mathbf{P}$; exchanging both vectors gives the other nodal plane. Strike, dip and rake follow the [Aki and Richards \(1980\)](#) convention. The trend of the slip vector of each plane is its azimuth, and its plunge is

$$\text{plunge} = \arcsin(\sin \lambda \sin \delta),$$

positive for slip with a reverse component (rake > 0) and negative for slip with a normal component (rake < 0).

Scalar seismic moment and magnitude. The scalar moment is computed following [Silver and Jordan \(1982\)](#), and the moment magnitude following [Hanks and Kanamori \(1979\)](#), with M_0 in dyn-cm:

$$M_0 = \sqrt{\frac{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}{2}}, \quad M_w = \frac{2}{3} \log_{10} M_0 - 10.7333$$

With AR input, M_0 is obtained from M_w by inverting this relation; with P input it is read from the mantissa and exponent.

CLVD ratio and Γ index. Both are computed from the eigenvalues of the deviatoric part of the moment tensor, $\lambda_P \leq \lambda_B \leq \lambda_T$, so they are not affected by the isotropic component:

$$f_{\text{clvd}} = -\frac{\lambda_B}{\max(|\lambda_T|, |\lambda_P|)}, \quad \Gamma = \frac{3\sqrt{6} \lambda_T \lambda_B \lambda_P}{(\lambda_T^2 + \lambda_B^2 + \lambda_P^2)^{3/2}}.$$

f_{clvd} ([Frohlich and Davis, 1999](#)) ranges from -0.5 to 0.5 and Gamma ([Kagan and Knopoff, 1985](#)) from -1 to 1 . Both are 0 for a double couple, positive for a tension-dominated CLVD, $(\lambda_T, \lambda_B, \lambda_P) \propto (2, -1, -1)$, and negative for a

compression-dominated one, $\propto (1, 1, -2)$. This is also the CLVD sign of the source-type diagram (Hudson et al., 1989; Vavryčuk, 2015), where positive values plot on the left:

Source		fclvd	Gamma	u_ Hudson	Position in the source-type diagram
Positive (tension-dominated)	CLVD	+0.5	+1	−1	left corner, “CLVD (+)”
Double couple		0	0	0	centre
Negative (compression-dominated)	CLVD	−0.5	−1	+1	right corner, “CLVD (−)”

Other definitions differ in sign or scale, so check the convention before comparing values: Giardini (1984) uses the opposite sign of fclvd (also the ϵ of Tape and Tape, 2012); the C_{CLVD} of Vavryčuk (2015) equals $2f_{\text{clvd}}$ for deviatoric tensors; and Γ is not linear in fclvd ($\Gamma \approx 2.6f_{\text{clvd}}$ for small values). Kagan and Knopoff (1985) caution that the accuracy of catalogue solutions is too low to interpret small CLVD values in terms of source complexity. Since they describe only the deviatoric part, for tensors with a dominant isotropic component they should be read together with fiso and the source-type diagram.

Isotropic component. The isotropic component is one third of the trace of the tensor, $\text{iso} = \text{tr}(\mathbf{M})/3$ (dyn-cm), and its ratio to the scalar moment is $f_{\text{iso}} = \text{iso}/M_0$. Global CMT solutions are constrained to be deviatoric, so for them these values are zero within the rounding of the input data.

Position on the diagrams. The coordinates on the classification diagram ($x_{\text{kav}}, y_{\text{kav}}$) are described in section 9, and those on the source-type diagram ($u_{\text{Hudson}}, v_{\text{Hudson}}$) in section 11.

Note on results from older versions. Since version 1.10, fclvd has the opposite sign to versions 1.8.1 to 1.9.1 (definition of Giardini, 1984), so old values must be multiplied by -1 ; it is also computed from the deviatoric eigenvalues, which changes it slightly (up to 0.005 in the Global CMT test data) when the input tensor has a non-zero trace. Before version 1.8.1 it was an absolute value (Frohlich and Apperson, 1992), and earlier versions computed the scalar moment as the average of the absolute largest and smallest eigenvalues (Dziewonski et al., 1981), so small differences in M_0 , m_{ant} and M_w are also expected when comparing with those versions.

9 Diagram

Frohlich and Apperson (1992) developed a ternary diagram to represent focal mechanisms based in the trigonometrical relation:

$$\sin^2 \iota_T + \sin^2 \iota_B + \sin^2 \iota_P = 1, \quad (9.1)$$

where ι_T , ι_B and ι_P are the plunges of the axis T, B and P of the SMT. This relation is true for three orthogonal axes.

As the equation of a sphere with radius unity is

$$x^2 + y^2 + z^2 = 1 \quad (9.2)$$

and all the plunge angles are positive between 0° and 90° , then the representation of a focal mechanism defined by these axes is equivalent to the projection of a point in an spherical octant over a planar surface. As a sphere is not a developable surface can not be projected onto a plane without distortion.

The projection used by Frohlich and Apperson (1992) represents this spherical octant as a triangle, which is equivalent to the gnomonic geographical projection. In this projection we define the angles

$$\psi = \arctan \left(\frac{\sin \iota_T}{\sin \iota_P} \right) - 45^\circ, \quad (9.3)$$

and

$$\iota_s = \arcsin \frac{1}{\sqrt{3}} = \arctan \frac{1}{\sqrt{2}} \approx 35.26^\circ, \quad (9.4)$$

which is the angle of the ternary symmetry axis with the orthogonal axes of the octant.

From these angles the x and y coordinates over the plane are given by:

$$x = \frac{\cos \iota_B \cdot \sin \psi}{\sin \iota_s \cdot \sin \iota_B + \cos \iota_s \cdot \cos \iota_B \cdot \cos \psi} \quad (9.5)$$

$$y = \frac{\cos \iota_s \cdot \sin \iota_B - \sin \iota_s \cdot \cos \iota_B \cdot \cos \psi}{\sin \iota_s \cdot \sin \iota_B + \cos \iota_s \cdot \cos \iota_B \cdot \cos \psi} \quad (9.6)$$

The angle ι_s in the projection is the angle of the plunges of the axes T, B and P for the focal mechanism that occupies the barycentre of the ternary diagram where $x = y = 0$.

As can be seen in Figure 15a this projection introduces remarkable distortions towards the extremes of the diagram. This distortions could difficult the study of some groups of focal mechanisms as the author pointed out (Frohlich, 2001). In order to avoid these distortions Kaverina et al. (1996)

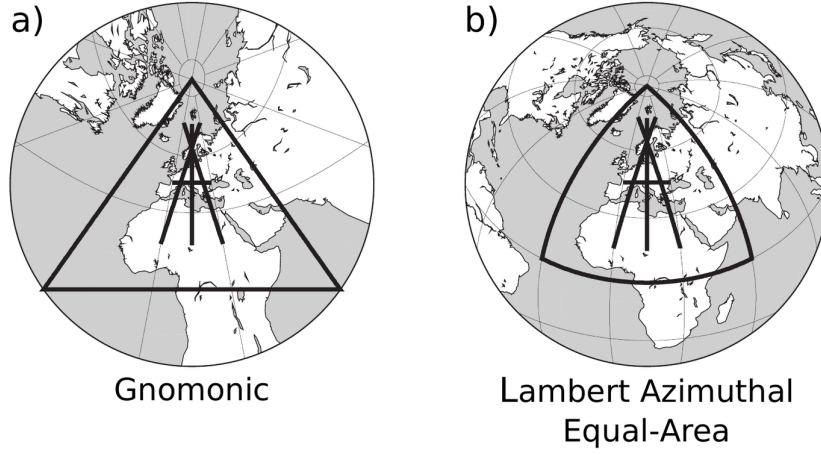


Figure 15: Representation of an hemisphere of the globe centred on Europe (45°N, 15°E) in the two different projections referenced in the text: a) Gnomonic projection; b) Lambert azimuthal equal-area. In thick lines are marked the spheric octant and some reference lines in order to compare both projections.

proposed the use of a projection capable of maintain the proportion of the areas equal, equivalent to the Lambert azimuthal equal-area projection (Figure 15b), which is the used in structural geology to generate the stereographic equal-area Schmidt stereonet. In this case the limits of the diagram are formed by great circles instead of straight lines forming a triangle (Figure 15).

If we take the sines of the plunges of the main axes ι_T , ι_B and ι_P :

$$z_T = \sin \iota_T, \quad z_P = \sin \iota_P, \quad z_B = \sin \iota_B \quad (9.7)$$

the length of the vector that connects the center of the diagram with the projected point is defined by

$$L = 2 \sin \left[\frac{1}{2} \cdot \arccos \left(\frac{z_T + z_P + z_B}{\sqrt{3}} \right) \right] \quad (9.8)$$

and the normalization factor is:

$$N = \sqrt{2 \cdot [(z_B - z_P)^2 + (z_B - z_T)^2 + (z_T - z_P)^2]} \quad (9.9)$$

The coordinates of the projected focal mechanisms in the plane are defined as follows (these equations correct a typo present in Kagan (2005)):

$$x = \sqrt{3} \cdot \frac{L}{N} \cdot (z_T - z_P) \quad (9.10)$$

$$y = \frac{L}{N} \cdot (2z_B - z_P - z_T) \quad (9.11)$$

10 Focal Mechanism Classification

In order to classify the focal mechanisms [Kagan \(2005\)](#) divided the octant in three areas (Figure 16b) corresponding to the three basic Andersonian regimes: normal, reverse and strike-slip. The dividing lines start at the diagram center and run through the middle of each great circle (dashed lines in Figure 16b). This classification, although simple and straightforward, is too basic sometimes for a detailed seismotectonic study.

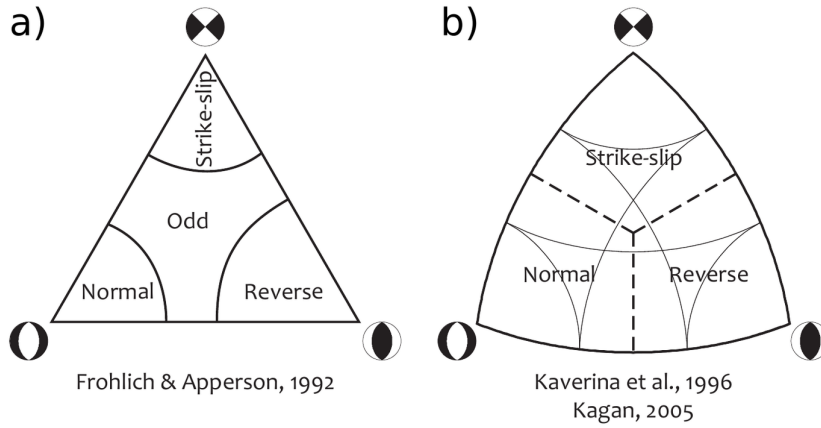


Figure 16: Focal mechanisms classification diagrams based on SMT axes plunges proposed by a) [Frohlich and Apperson \(1992\)](#) and b) [Kagan \(2005\)](#).

From the seismotectonic point of view an earthquake reflects the brittle lithospheric strain produced by tectonic processes. The vast majority of earthquakes are produced by displacements on faults (some other processes as deep volume changes or cave collapses can produce seismic signals too), and they are interpreted usually as fault ruptures. Following this reasoning the most appropriate classification for the focal mechanism should be done by means of the slip vector on the fault, given by the rake angle. This classification can be done following one of the proposed conventions ([Rickard, 1972](#); [Aki and Richards, 1980](#); [Angelier, 1994](#)). As has been mentioned before, the problem is the duplicity of data due to the two nodal planes solution of the focal mechanism.

An extended discussion on the relation among the different focal mechanism parameters can be found on [Célérrier \(2010\)](#); the author also points out the difficulty on interpreting the stress state from the focal mechanisms, specially where oblique faulting takes place and reactivation of inherited structures is possible ([Wyss et al., 1992](#)).

However the SMT can be interpreted as a seismic strain tensor ([Kostrov, 1974](#)) where the main axes are equivalent to the SMT main axes ([Wyss et al., 1992](#)). With this relation in mind we can interpret the populations of focal mechanisms as representations of strains in different tectonic settings. Oblique slip focal mechanisms with normal component can be interpreted as transtensional strain, while oblique slip focal mechanisms with reverse component can be interpreted as transpressional.

In order to count with a classification more detailed than the one of [Kagan \(2005\)](#) I decided to classify the focal mechanism in a series of fields that include the oblique slip regimes ([Álvarez-Gómez, 2009](#)). This approximation is similar to the [Johnston et al. \(1994\)](#) classification; with 7 classes of earthquakes: 1) Normal; 2) Normal - Strike-slip; 3) Strike-slip - Normal; 4) Strike-slip; 5) Strike-slip - Reverse; 6) Reverse - Strike-slip and 7) Reverse. The resulting diagram incorporating this 7 fields classification to the Kaverina projection is shown in Figure 17.

The algorithm to classify the focal mechanisms based on the SMT main axes plunges is the shown in Figure 1.

When one of the three main SMT axes is greater or equal than 67.5° (resulted from $3 \cdot \frac{\pi/2}{4}$) we obtain a "pure" Andersonian tectonic regime. From the pure normal or reverse faulting to the strike-slip faulting there is a range of oblique faulting types with more or less relevance of the horizontal component. On the other hand between the pure normal and pure reverse faulting there is a permutation of the T and P axes in the vertical position, while the B axis remains horizontal. The transition takes place when the P and T axes plunge equally and a vertical nodal plane is present.

In the following section I compare the results of both classifications, the one based in the main axes plunges of the moment tensor and the one based on the fault rakes.

10.1 Comparison between rake-based and SMT axes-based classifications

In order to classify the nodal planes according to its rakes I adopted a convention hybrid between the [Rickard \(1972\)](#) and [Aki and Richards \(1980\)](#) conventions. The focal mechanism rakes are usually given with the [Aki and Richards \(1980\)](#) convention, while from a geological point of view the

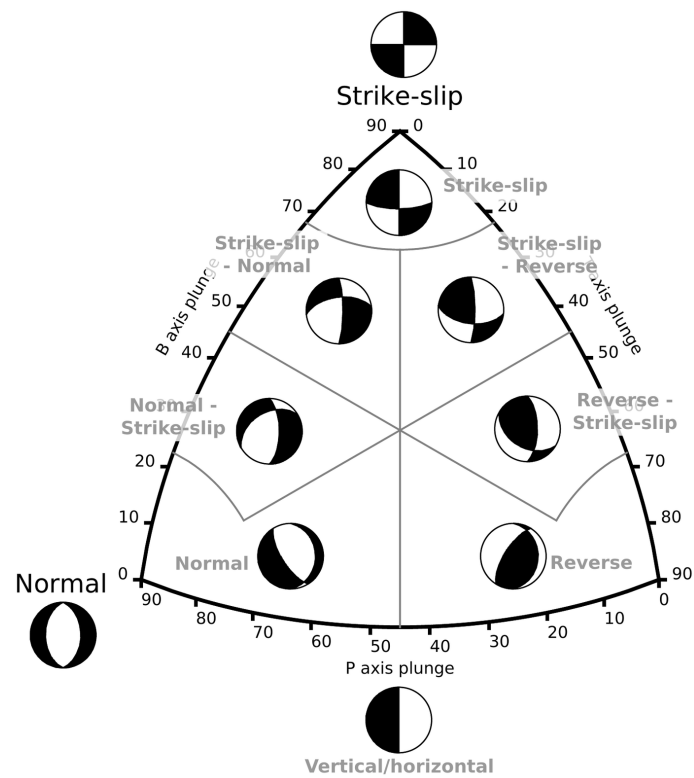


Figure 17: Focal mechanism classification diagram. N) Normal; N-SS) Normal - Strike-slip; SS-N) Strike-slip - Normal; SS) Strike-slip; SS-R) Strike-slip - Reverse; R-SS) Reverse - strike-slip and R) Reverse.

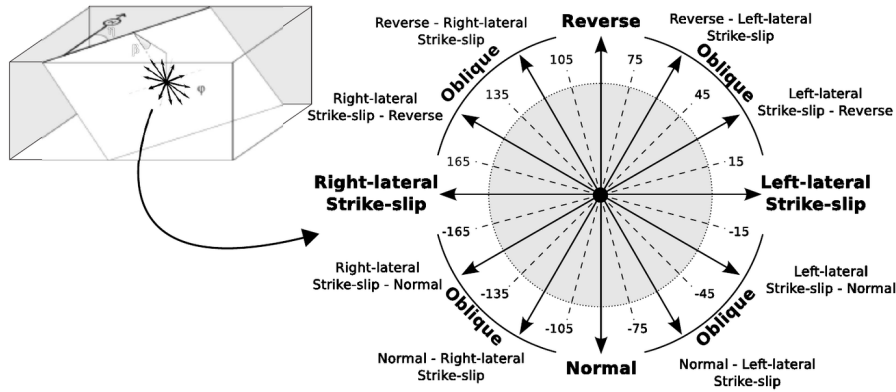


Figure 18: Diagram showing the convention used in this work for the rake classification of faults or focal mechanism nodal planes. Inside the grey circle the equivalent SMT axes-based classification is shown (Figure 17).

[Rickard \(1972\)](#) convention is more used. I define 7 equivalent classes to those used in the SMT axes-based classifications (Figure 18).

Each one of the nodal planes of the focal mechanisms from the Global CMT catalog ([Ekström et al., 2012](#)) is classified according to the rake classification shown in Figure 18. From the analysis of the focal mechanism alone we cannot choose unequivocally the fault plane responsible of the earthquake. To analyze the degree of uncertainty I show in Figure 19a the relative frequency of relations between nodal planes rupture types. For example, it can be seen that when one nodal plane is classified as normal, the other can be normal as well, or can present some strike-slip component, but it can not present reverse component; the same can be said for the reverse nodal planes. When one nodal plane is pure strike-slip, the other can present dip-slip component too, ranging from normal with strike-slip component (N-SS) to reverse with strike-slip component (R-SS), but being more frequent the relations between nodal planes of mainly strike-slip type.

Comparing the focal mechanism classified with the SMT axes and the nodal planes classified with the rakes (Figure 19b) a similar picture as the described above can be seen. The percentages of relations change sensibly, but the common picture is the same. When the focal mechanism is classified as normal type we can expect that both nodal planes are of normal type or normal with some strike-slip component. Similarly when the focal mechanism is classified as reverse, both nodal planes are mainly of the reverse type too. For the case of the strike-slip focal mechanism almost all the nodal planes are of strike-slip type too.

We can conclude that there is not much difference between the results obtained from the rake classification of the nodal planes and the classifica-

tion of the focal mechanism attending to the SMT axes plunges. The focal mechanism classification based on the SMT axes has the advantage of the univocality of its classification, in contrast with the duplicity of data and uncertainty related to the nodal planes rake classification.

11 The source-type diagram

The classification diagram describes the double-couple part of the moment tensor. To analyse the non-double-couple components, FMC also plots the source-type diagram of [Hudson et al. \(1989\)](#), in the skewed-diamond form described by [Vavryčuk \(2015\)](#) (flag -pd). The eigenvalues $\lambda_1 \leq \lambda_2 \leq \lambda_3$ are normalised by the largest absolute eigenvalue, $\hat{\lambda}_i = \lambda_i / \max_j |\lambda_j|$, and the coordinates are

$$u = -\frac{2}{3} (\hat{\lambda}_1 + \hat{\lambda}_3 - 2\hat{\lambda}_2), \quad v = \frac{1}{3} (\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3).$$

The vertical coordinate v measures the isotropic component: pure explosions plot at the top vertex ($v = 1$) and pure implosions at the bottom one ($v = -1$). The horizontal coordinate u measures the CLVD component: a pure double couple plots at the centre, and pure CLVD sources at $u = \pm 1$ on the horizontal axis. Because u is minus the CLVD coordinate of the standard decomposition ([Vavryčuk, 2015](#), eqs. 7 and 42), positive CLVDs (tension-dominated) plot at the left corner ($u = -1$, labelled “CLVD (+)”) and negative CLVDs (compression-dominated) at the right corner ($u = +1$, labelled “CLVD (-)”). For deviatoric tensors $v = 0$ and $u = -2f_{\text{clvd}}$, so events with positive f_{clvd} plot on the left and events with negative f_{clvd} on the right (section 8). The dashed diagonal line of the diagram joins the two lateral vertices, at $(\pm 4/3, \pm 1/3)$.

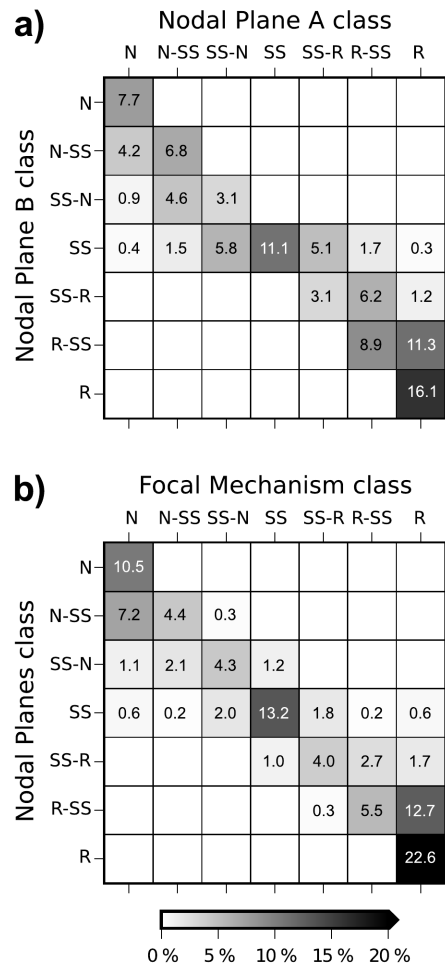


Figure 19: Heatmaps of the proportion of relations between a) both nodal planes rupture type based on the rake classification, and b) between the focal mechanism classification and the nodal planes rupture types rake classification. The data used is the entire Global CMT catalog (Ekström et al., 2012). Numbers show percentages over the entire catalogue.

References

- Aki, K., Richards, P., 1980. Quantitative Seismology, Theory and Methods. volume I and II. W.H. Freeman, San Francisco.
- Álvarez-Gómez, J., 2009. Tectónica Activa y Geodinámica en el Norte de Centroamérica. Ph.D. thesis. Universidad Complutense de Madrid. Madrid.
- Angelier, J., 1994. Palaeostress analysis of small-scale brittle structures, in: Hancock, P. (Ed.), Continental Deformation. Pergamon Press, p. 421.
- Benioff, H., 1954. Orogenesis and deep crustal structure — additional evidence from seismology. Geological Society of America Bulletin 65, 385–400.
- Borges, J., Fitas, A., Bezzeghoud, M., Teves-Costa, P., 2001. Seismotectonics of portugal and its adjacent atlantic area. Tectonophysics 331, 373–387.
- Bullen, K., Bolt, B., 1985. An Introduction to the Theory of Seismology. 4th ed., Cambridge University Press, Cambridge.
- Byerly, P., 1928. The nature of the first motion in the chilean earthquake of november 11, 1922. American Journal of Science, Series 5 16, 232–236.
- Byerly, P., 1938. The earthquake of july 6, 1934: amplitudes and first motion. Bulletin of the Seismological Society of America 28, 1–13.
- Célérier, B., 2010. Remarks on the relationship between the tectonic regime, the rake of the slip vectors, the dip of the nodal planes, and the plunges of the p, b, and t axes of earthquake focal mechanisms. Tectonophysics 482, 42–49. doi:[10.1016/j.tecto.2009.03.006](https://doi.org/10.1016/j.tecto.2009.03.006).
- Davis, G., Reynolds, S., 1996. Structural Geology of Rocks and Regions. 2nd ed., Wiley, New York.
- Dziewonski, A., Chou, T., Woodhouse, J., 1981. Determination of earthquake source parameters from waveform data for studies of global and regional seismicity. Journal of Geophysical Research 86, 2825–2852.
- Ekström, G., Nettles, M., Dziewoński, A., 2012. The global cmt project 2004–2010: centroid-moment tensors for 13,017 earthquakes. Physics of the Earth and Planetary Interiors 200–201, 1–9.
- Frohlich, C., 2001. Display and quantitative assessment of distributions of earthquake focal mechanisms. Geophysical Journal International 144, 300–308. doi:[10.1046/j.1365-246x.2001.00341.x](https://doi.org/10.1046/j.1365-246x.2001.00341.x).

- Frohlich, C., Apperson, K., 1992. Earthquake focal mechanisms, moment tensors, and the consistency of seismic activity near plate boundaries. *Tectonics* 11, 279–296.
- Frohlich, C., Coffin, M., Massell, C., Mann, P., Schuur, C., Davis, S., Jones, T., Karner, G., 1997. Constraints on macquarie ridge tectonics provided by harvard focal mechanisms and teleseismic earthquake locations. *Journal of Geophysical Research* 102, 5029–5041.
- Frohlich, C., Davis, S., 1999. How well constrained are well-constrained t , b , and p axes in moment tensor catalogs? *Journal of Geophysical Research* 104, 4901–4910.
- Gasperini, P., Vannucci, G., 2003. Fpspack: a package of fortran subroutines to manage earthquake focal mechanism data. *Computers & Geosciences* 29, 893–901.
- Gephart, J., 1990. Stress and the direction of slip on fault planes. *Tectonics* 9, 845–858.
- Giardini, D., 1984. Systematic analysis of deep seismicity: 200 centroid-moment tensor solutions for earthquakes between 1977 and 1980. *Geophysical Journal of the Royal Astronomical Society* 77, 883–914.
- Hanks, T., Kanamori, H., 1979. A moment magnitude scale. *Journal of Geophysical Research* 84, 2348–2350.
- Hess, H., 1962. History of ocean basins. *Petrologic Studies* 4, 599–620.
- Holmes, A., 1931. Radioactivity and earth movements. *Nature* 128, 496.
- Honda, H., Masatuke, A., 1952. On the mechanism of the earthquakes and the stresses producing them in japan and its vicinity. *Science Reports of the Tohoku University* 4.
- Hudson, J., Pearce, R., Rogers, R., 1989. Source type plot for inversion of the moment tensor. *Journal of Geophysical Research* 94, 765–774.
- Igarashi, T., Matsuzawa, T., Umino, N., Hasegawa, A., 2001. Spatial distribution of focal mechanisms for interplate and intraplate earthquakes associated with the subducting pacific plate beneath the northeastern japan arc: a triple-planed deep seismic zone. *Journal of Geophysical Research* 106, 2177–2191.
- Isacks, B., Molnar, P., 1969. Mantle earthquake mechanisms and the sinking of the lithosphere. *Nature* 223, 1121–1124.
- Isacks, B., Oliver, J., Sykes, L., 1968. Seismology and the new global tectonics. *Journal of Geophysical Research* 73, 5855–5899.

- Johnston, A., Coppersmith, K., Kanter, L., Cornell, C., 1994. The Earthquakes of Stable Continental Regions. Volume 1: Assessment of Large Earthquake Potential. Technical Report. Electric Power Research Institute.
- Kagan, Y., 2005. Double-couple earthquake focal mechanism: random rotation and display. *Geophysical Journal International* 163, 1065–1072.
- Kagan, Y., Knopoff, L., 1985. The first-order statistical moment of the seismic moment tensor. *Geophysical Journal of the Royal Astronomical Society* 81, 429–444.
- Kaverina, A., Lander, A., Prozorov, A., 1996. Global creepex distribution and its relation to earthquake-source geometry and tectonic origin. *Geophysical Journal International* 125, 249–265.
- Kita, S., Okada, T., Nakajima, J., Matsuzawa, T., Hasegawa, A., 2006. Existence of a seismic belt in the upper plane of the double seismic zone extending in the along-arc direction at depths of 70–100 km beneath ne japan. *Geophysical Research Letters* 33.
- Knopoff, L., 1961. Analytical calculation of the fault-plane problem. *Publications of the Dominion Observatory (Ottawa)* 24, 309–315.
- Koning, L., 1941. On the mechanism of deep focus earthquakes. *Gerlands Beiträge zur Geophysik* 58, 159–197.
- Kostrov, V., 1974. Seismic moment and energy of earthquakes, and seismic flow of rock. *Izvestiya, Physics of the Solid Earth* 1, 23–40.
- Langston, C., Helmberger, D., 1975. A procedure for modelling shallow dislocation sources. *Geophysical Journal of the Royal Astronomical Society* 42, 117–130.
- Le Pichon, X., 1968. Sea-floor spreading and continental drift. *Journal of Geophysical Research* 73, 3661–3697.
- McKenzie, D., 1969. The relation between fault plane solutions for earthquakes and the directions of the principal stresses. *Bulletin of the Seismological Society of America* 59, 591–601.
- Michael, A., 1987. Use of focal mechanisms to determine stress: a control study. *Journal of Geophysical Research* 92, 357–368.
- Morgan, W., 1968. Rises, trenches, great faults, and crustal blocks. *Journal of Geophysical Research* 73, 1959–1982.

- Okada, Y., 1992. Internal deformation due to shear and tensile faults in a half-space. *Bulletin of the Seismological Society of America* 82, 1018–1040.
- Pollard, D., Fletcher, R., 2005. *Fundamentals of Structural Geology*. Cambridge University Press, Cambridge.
- Ramsay, J., Huber, M., 1997. *The Techniques of Modern Structural Geology*, Vol. 2: Folds and Fractures. 5th ed., Academic Press, London.
- Ratchkovski, N., Hansen, R., 2002. New evidence for segmentation of the alaska subduction zone. *Bulletin of the Seismological Society of America* 92, 1754–1765.
- Rickard, M., 1972. Fault classification: discussion. *Geological Society of America Bulletin* 83, 2545–2546.
- Scheidegger, A., 1957. The geometrical representation of fault-plane solutions of earthquakes. *Bulletin of the Seismological Society of America* 47, 89–110.
- Scholz, C., 2002. *The Mechanics of Earthquakes and Faulting*. 2nd ed., Cambridge University Press, Cambridge.
- Schwinner, R., 1919. *Vulkanismus und Gebirgsbildung: ein Versuch*. Reimer.
- Serpelloni, E., Vannucci, G., Pondrelli, S., Argnani, A., Casula, G., Anzidei, M., Baldi, P., Gasperini, P., 2007. Kinematics of the western africa–eurasia plate boundary from focal mechanisms and gps data. *Geophysical Journal International* 169, 1180–1200.
- Silver, P., Jordan, T., 1982. Optimal estimation of scalar seismic moment. *Geophysical Journal of the Royal Astronomical Society* 70, 755–787.
- Sykes, L., 1967. Mechanism of earthquakes and nature of faulting on the mid-oceanic ridges. *Journal of Geophysical Research* 72, 2131–2153.
- Tape, W., Tape, C., 2012. A geometric setting for moment tensors. *Geophysical Journal International* 190, 476–498.
- Taylor, F., 1910. Bearing of the tertiary mountain belt on the origin of the earth's plan. *Geological Society of America Bulletin* 21, 179–226.
- Udías, A., 1989. Parámetros del foco de los terremotos. *Física de la Tierra* 1, 87–104.
- Vannucci, G., Gasperini, P., 2003. A database of revised fault plane solutions for italy and surrounding regions. *Computers & Geosciences* 29, 903–909.

- Vavryčuk, V., 2015. Moment tensor decompositions revisited. *Journal of Seismology* 19, 231–252. doi:[10.1007/s10950-014-9463-y](https://doi.org/10.1007/s10950-014-9463-y).
- Vine, F., 1966. Spreading of the ocean floor: new evidence. *Science* 154, 1405–1415.
- Wegener, A., 1912. Die entstehung der kontinente. *Geologische Rundschau* 3, 276–292.
- Wessel, P., Smith, W., Scharroo, R., Luis, J., Wobbe, F., 2013. Generic mapping tools: improved version released. *Eos, Transactions American Geophysical Union* 94, 409–410.
- Wyss, M., Liang, B., Tanigawa, W., Wu, X., 1992. Comparison of orientations of stress and strain tensors based on fault plane solutions in kaoiki, hawaii. *Journal of Geophysical Research* 97, 4769–4790.